

HEALTHCARE DATA CLIENT

Findings & Recommendations

Healthcare Data Quality Automation and Exception Reporting

Document Control	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Batch Scale	Approximately 250,000 records per batch
Observed Exception Rate	2%
Review Turnaround	Reduced from 6 hours to 2.5 hours per file

Evidence boundary: Only findings supported by the confirmed engagement facts are presented as historical results. Recommendations below are portfolio recommendations derived from the validated workflow and controls, not claimed as verbatim historical client recommendations.

1. Executive Findings

Finding	Evidence	Business Interpretation
High-volume validation was operationalized	Approximately 250,000 records were processed per batch.	The solution supported repeatable review at a scale that made complete manual record-by-record validation inefficient.
Exceptions were made visible	Observed record-level Exception Rate was 2%.	Records failing one or more validation rules could be isolated for targeted review.
Exception review became traceable	Failed records were separated, highlighted and assigned a reason code or equivalent explanation.	Reviewers could identify why a record required attention without rediscovering the validation failure manually.
Review turnaround improved materially	Review time reduced from 6 hours to 2.5 hours per file.	The automated validation and targeted exception workflow reduced turnaround by 3.5 hours per file.
Turnaround reduction was substantial	Derived reduction: approximately 58.3%.	The improvement applies specifically to review turnaround, not to labor cost, clinical quality or compliance performance.
Core result: The engagement combined automated validation with targeted exception review, reducing review turnaround while preserving record-level failure visibility.		

2. Data Quality Findings

Area	Finding	What Can Be Concluded
Validation coverage	The implemented process checked duplicates, missing values, invalid dates, incorrect formats, invalid IDs, out-of-range values, inconsistent codes and cross-field mismatches.	The solution addressed multiple dimensions of data quality rather than relying on a single validation check.
Record-level exceptions	A record failing one or more applicable rules was treated as an exception.	The 2% Exception Rate is based on affected records, not total rule failures.
Multiple failures	One record could fail more than one rule.	Rule-level failure analysis must remain separate from record-level Exception Rate to prevent double-counting.
Exception classification	Reason codes or equivalent explanations were attached to failed records.	Exception outputs supported more focused analyst review.
Historical distribution	Per-rule and per-domain exception distributions are not retained.	No category or healthcare domain should be presented as the largest source of exceptions.

3. Process Findings

Process Area	Finding
Input handling	Excel, CSV and SQL/database sources were within the confirmed engagement scope.
Validation workflow	The confirmed flow was Ingest > Validate > Apply Data Quality Rules > Flag Exceptions > Assign Reason Codes > Generate Exception Output / Report > Review.
Exception handling	Failed records were routed for review rather than silently corrected or discarded.
Reporting	Power BI provided data-quality and exception-reporting visibility.
Quality assurance	Requirements, business rules, testing and documentation formed part of the engagement.

4. Business Impact

Impact Measure	Before	After / Result	Change
Review turnaround per file	6 hours	2.5 hours	3.5 hours less
Review turnaround reduction	Baseline	Post-solution	Approximately 58.3% derived
Exception visibility	Manual review context	2% record-level Exception Rate surfaced	Targeted exception population
Processing scale	High-volume healthcare data	Approximately 250,000 records per batch supported	Repeatable validation at batch scale

Impact boundary: The engagement evidence supports turnaround improvement and exception visibility. It does not support claims of clinical-outcome improvement, patient-safety improvement, regulatory compliance improvement, labor-cost savings or a specific reduction in healthcare errors.

5. Portfolio Recommendations

Priority	Recommendation	Rationale	Evidence Status
High	Maintain a governed rule catalogue with rule IDs, logic, applicable fields, parameters and reason classifications.	Keeps validation repeatable, testable and understandable as source rules evolve.	Portfolio recommendation
High	Preserve separate record-level and rule-level validation outputs.	Prevents multi-failure records from inflating Exception Rate and supports deeper reason analysis.	Portfolio recommendation
High	Reconcile processed, exception and validation-failure populations for every batch.	Protects KPI integrity and improves confidence in downstream Power BI reporting.	Portfolio recommendation
High	Retain source-record and batch traceability through exception review.	Allows analysts to investigate flagged records without losing lineage to the processed source.	Portfolio recommendation
Medium	Monitor exception patterns by governed reason code when sufficient historical data is available.	Can identify recurring validation issues without assuming unsupported historical distributions.	Portfolio recommendation
Medium	Use regression tests when rule parameters, formats, ranges, code sets or cross-field relationships change.	Reduces the risk that rule changes unintentionally alter exception classification.	Portfolio recommendation
Medium	Keep Power BI measures aligned with the governed validation outputs.	Prevents reporting logic from diverging from automated validation logic.	Portfolio recommendation
Medium	Keep public portfolio outputs free of PHI and confidential client values.	Protects healthcare-data confidentiality while preserving analytical credibility.	Portfolio requirement

6. Recommendation Traceability

Recommendation Theme	Supporting Artifact
Rule governance	Data Quality Rule Catalogue
Record vs failure grain	Data Dictionary; KPI Catalogue; Data Model & DAX Measures
Batch reconciliation	Automated Validation Workflow; Exception Analysis
Reason-code analysis	Exception Analysis; Power BI Requirements
Regression testing	User Stories & Acceptance Criteria; Data Quality Rule Catalogue
Reporting consistency	Power BI Requirements; Data Model & DAX Measures
Confidentiality	BRD; Data Dictionary; Power BI Requirements

7. What the Evidence Does Not Support

Unsupported Claim	Reason
A specific validation category caused most exceptions	Historical per-rule distribution is not retained.
A specific healthcare domain had the highest exception rate	Historical domain-level distribution is not retained.
The 2% rate was an acceptable threshold or SLA	2% is an observed result, not a confirmed target.
Automation improved clinical outcomes or patient safety	No such outcome evidence was provided.
Automation produced a specific cost saving	No cost baseline or labor-cost calculation was provided.
The remaining 2.5 hours consisted of specific manual activities	Detailed composition of the post-solution review time is not retained.
Specific healthcare code sets or clinical validation logic were used	Exact field-level healthcare rules are not retained.

8. Interview-Ready Summary

The healthcare engagement focused on improving data-quality review for high-volume files. I worked with approximately 250,000 records per batch and implemented automated checks across duplicates, missing values, dates, formats, identifiers, ranges, codes and cross-field mismatches. Records failing validation were separated for review and given reason information, which surfaced a 2% record-level exception rate. The targeted workflow reduced review turnaround from approximately 6 hours to 2.5 hours per file, a reduction of 3.5 hours or about 58.3%. In the portfolio, I keep record-level exceptions separate from rule-level failures so that records with multiple validation issues do not inflate the Exception Rate.

9. Final Conclusion

Conclusion: The strongest evidence from this engagement is not a fabricated healthcare insight. It is the combination of scale, governed validation, traceable exception handling and measurable review-turnaround improvement. That makes the case study defensible as a Data Analyst solution focused on data quality, automation and operational review efficiency.